

DAX FOR EXCEL - COMPLETE GUIDE

1. INTRODUCTION TO DAX

WHAT IS DAX?

DAX stands for **Data Analysis Expressions**. It is a formula language used in **Power Pivot** (Excel) and **Power BI** to perform calculations on data models.

WHY DAX WAS INTRODUCED?

- Excel formulas are limited to single tables.
- DAX works on Relationships between tables.
- Handles large data efficiently.
- Enables advanced analytics & KPIs.
- Used to create Measures, Calculated Columns and Calculated Tables.



DAX VS EXCEL FORMULAS

Feature	Excel Formula	DAX
Works On	Single Table	Multiple Related Tables
Calculation	Cell by Cell	Column/Model Based
Performance	Slower on Big Data	Very Fast (In-Memory)
Logic	Limited	Advanced & Powerful
Purpose	Data Entry & Basic Calculations	Business Intelligence & Analytics

WHERE DAX WORKS?



POWER PIVOT
(EXCEL)

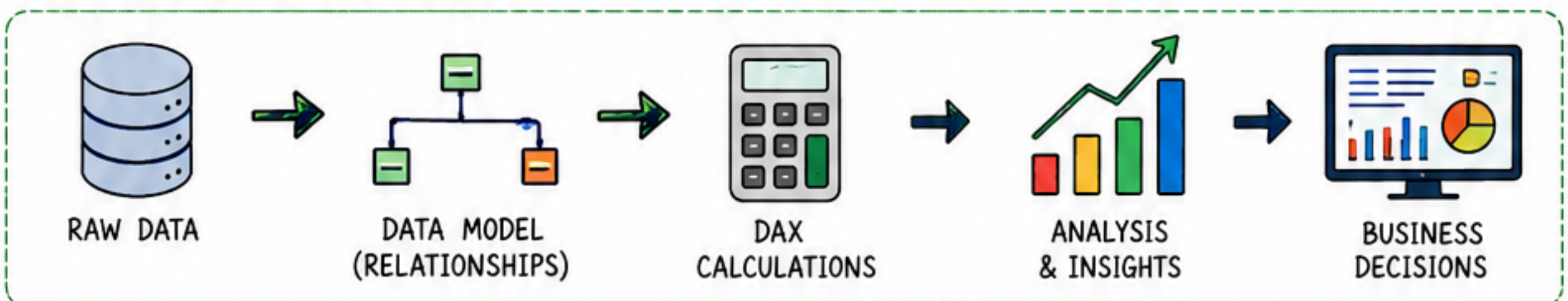


POWER BI

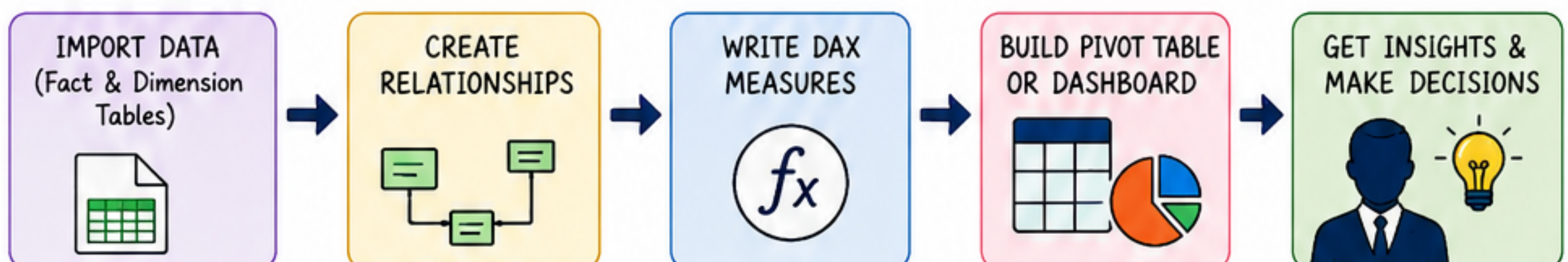
WHY EVERY DATA ANALYST SHOULD LEARN DAX?

- ✓ Create powerful business metrics (KPIs).
- ✓ Analyse large datasets easily.
- ✓ Build interactive dashboards.
- ✓ Get high paying roles in Data & BI.
- ✓ Crucial for Power BI & Advanced Excel.

REAL BUSINESS SCENARIO



DAX WORKFLOW



★ KEY TAKEAWAYS

- ✓ DAX is the backbone of Power Pivot & Power BI.
- ✓ It helps in creating advanced calculations and KPIs.
- ✓ Works on data models, not just single tables.
- ✓ Essential skill for every Data Analyst.

“DAX turns data into meaningful business insights.”





2. MEASURES vs CALCULATED COLUMNS vs CALCULATED TABLES

MEASURE



- Calculated on the fly.
- Stored in the Data Model.
- Changes with Filter Context.
- Used in Pivot Tables & Reports.
- Icon:

Example

Total Sales = SUM(Sales[Amount])

CALCULATED COLUMN



- Calculated row by row.
- Stored in the Data Model.
- Changes with Row Context.
- Used like a normal column.
- Icon:

Example

Discount % =
Sales[Discount] / Sales[Amount]

CALCULATED TABLE



- Created using DAX expression.
- Stored in the Data Model.
- Independent table.
- Used for advanced analysis.
- Icon:

Example

High Sales Customers =
FILTER(VALUES(Customer[Name]),
[Total Sales] > 100000)

3. ROW CONTEXT vs FILTER CONTEXT



ROW CONTEXT

- Refers to the current row.
- Created by Calculated Columns or Iterators (e.g., SUMX, FILTER).
- Knows "which row" is being processed.

Example: In calculated column, we are looking at one row at a time.

VS



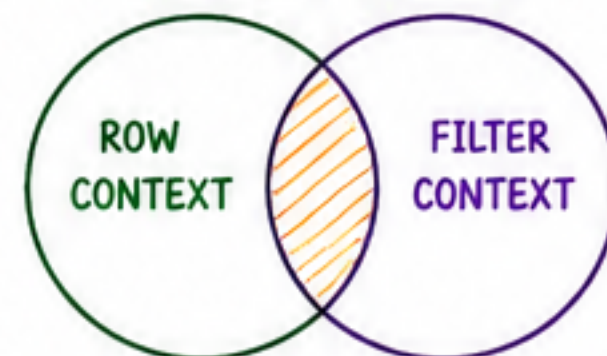
FILTER CONTEXT

- Refers to the filters applied.
- Created by reports, slicers, CALCULATE, or filter functions.
- Determines "which data is visible".

Example: When we apply Year = 2024 in a slicer, data is filtered.

EVALUATION CONTEXT

DAX expression is evaluated based on Row Context, Filter Context, or both.



EVALUATION CONTEXT

4. AGGREGATION FUNCTIONS

Function	Syntax	Description	Example	Use Case
SUM	SUM(Table[Column])	Adds all values.	Total Sales = SUM(Sales[Amount])	Total Revenue
SUMX	SUMX(Table, Expression)	Adds values of an expression row by row.	Total Sales = SUMX(Sales, Sales[Qty] * Sales[Price])	Total Sales (Qty x Price)
AVERAGE	AVERAGE(Table[Column])	Average of values.	Avg Price = AVERAGE(Products[Price])	Average Price
AVERAGEX	AVERAGEX(Table, Expression)	Average of an expression row by row.	Avg Order Value = AVERAGEX(Orders, Orders[Amount])	Average Order Value
MIN / MAX	MIN(Table[Column]) MAX(Table[Column])	Smallest / Largest value.	Max Sales = MAX(Sales[Amount])	Min/Max Sales
COUNT	COUNT(Table[Column])	Counts numeric values.	Total Orders = COUNT(Orders[OrderID])	Total Orders
COUNTA	COUNTA(Table[Column])	Counts non-blank values.	Total Customers = COUNTA(Customer[Name])	Total Customers
COUNTRWS	COUNTRWS(Table)	Counts total rows.	Total Rows = COUNTRWS(Sales)	Total Transactions
DISTINCTCOUNT	DISTINCTCOUNT(Table[Column])	Counts unique values.	Unique Customers = DISTINCTCOUNT(Customer[ID])	Unique Customers
DIVIDE	DIVIDE(Numerator, Denominator, [Alternate Result])	Safe division (handles /0).	Margin % = DIVIDE([Profit], [Sales], 0)	Margins, Ratios
COALESCE	COALESCE(Expression, Alt)	Returns first non-blank.	Region = COALESCE(Customer[Region], "Not Available")	Handling Blank Values



KEY TAKEAWAYS

- ✓ Measures change with filters.
- ✓ Calculated Columns do not change with filters.
- ✓ Evaluating context is the key to writing DAX.
- ✓ Aggregation functions are the building blocks of analytics.

QUICK EXAMPLE

Total Sales = SUM(Sales[Amount])
Total Cost = SUM(Sales[Cost])
Total Profit = [Total Sales] - [Total Cost]
Profit % = DIVIDE([Total Profit], [Total Sales], 0)

BUSINESS USE CASE

Helps in building KPIs like Revenue, Profit, Margin, Growth %, Contribution % for decision making.



REMEMBER

DAX is not just a formula language, it is a **thinking language**.
It helps you ask the right business questions and get meaningful answers!

“ Learn DAX deeply,
Your Analytics career
will rise strongly. ”

- Tech With Shivansh



5. LOGICAL FUNCTIONS

Function	Syntax	Description	Example	Use Case
IF	IF(Condition, True Result, False Result)	Returns a value based on condition.	IF([Sales] > 1000, "High", "Low")	Classify High/Low Sales
SWITCH	SWITCH(Expression, Value, Result, ..., Else)	Checks expression against multiple values.	SWITCH(TRUE(), [Profit] > 0, "Profit", [Profit] = 0, "Break Even", "Loss")	Multi-condition logic
AND	AND(Condition1, Condition2, ...)	Returns TRUE if all conditions are TRUE.	AND([Sales] > 0, [Profit] > 0)	Multiple conditions must be true
OR	OR(Condition1, Condition2, ...)	Returns TRUE if any condition is TRUE.	OR([Sales] > 0, [Profit] > 0)	At least one condition true
NOT	NOT(Condition)	Reverses TRUE to FALSE and vice versa.	NOT(ISBLANK([Sales]))	Check Not Blank
ISBLANK	ISBLANK(Expression)	Checks if value is blank.	ISBLANK([Sales])	Data completeness
BLANK	BLANK()	Returns a blank value.	IF([Sales]=0, BLANK(), [Sales])	Hide zeros in reports

EXAMPLE - LOGICAL USE CASE

Scenario: Mark Customers as Active if they have purchased in the last 90 days.

Active Customer =
IF (
DATEDIFF (Customer[Last Purchase Date],
TODAY(), DAY) <= 90,
"Active", "Inactive"
)



★ KEY TAKEAWAY

- ✓ Logical functions help in decision making.
- ✓ SWITCH is cleaner than nested IF.
- ✓ ISBLANK and NOT are very useful in reports.



6. FILTER MODIFIER FUNCTIONS (MOST IMPORTANT)

Function	Syntax	Description	Example	Use Case
CALCULATE	CALCULATE(Expression, Filter1, Filter2, ...)	Evaluates an expression in a modified filter context.	CALCULATE(SUM(Sales[Amount]), Sales[Region] = "North")	Change filter context
FILTER	FILTER(Table, Condition)	Returns a table with rows that meet condition.	FILTER(Sales, Sales[Amount] > 1000)	Filter table manually
ALL	ALL(Table/Column)	Removes all filters from table/column.	CALCULATE([Total Sales], ALL(Sales[Date]))	Ignore date filter
ALLEXCEPT	ALLEXCEPT(Table, Column1, Column2, ...)	Removes all filters except selected columns.	ALLEXCEPT(Sales, Sales[Region])	Keep specific filters
ALLSELECTED	ALLSELECTED(Table/Column)	Removes filters but keeps external filters (Slicers).	CALCULATE([Total Sales], ALLSELECTED(Sales[Category]))	Visual Totals
REMOVEFILTERS	REMOVEFILTERS(Table/Column)	Same as ALL, used to remove filters.	CALCULATE([Total Sales], REMOVEFILTERS(Sales[Date]))	Clear filters
KEEPFILTERS	KEEPFILTERS(Condition)	Adds filter without overwriting existing filters.	CALCULATE([Total Sales], KEEPFILTERS(Sales[Region] = "North"))	Add filter respectively
CROSSFILTER	CROSSFILTER(Column1, Column2, Direction)	Changes filter direction between tables.	CROSSFILTER(Date[Date ID], Sales[Date ID], BOTH)	Change filter behavior
USERELATIONSHIP	USERELATIONSHIP(Column1, Column2)	Uses an inactive relationship in calculation.	CALCULATE([Total Sales], USERELATIONSHIP(Date[Date], Sales[Ship Date]))	Use inactive relationship

EXAMPLE - CALCULATE + FILTER

Total Sales of High Value Transactions (Amount > 1000)

High Value Sales =
CALCULATE (
SUM (Sales[Amount]),
FILTER (Sales, Sales[Amount] > 1000)
)



BUSINESS USE CASES

- ✓ Apply dynamic filters.
- ✓ Calculate KPIs under different conditions.
- ✓ Create comparisons across regions, time periods, etc.
- ✓ Build interactive dashboards with slicers.



7. RELATIONSHIP & LOOKUP FUNCTIONS

Function	Syntax	Description	Example
RELATED	RELATED(Column)	Returns a related column value from another table.	RELATED(Customer[Customer Name])
RELATEDTABLE	RELATEDTABLE(Table)	Returns a table of related rows.	RELATEDTABLE(Sales)
LOOKUPVALUE	LOOKUPVALUE(Result_Column, Search_Column, Search_Value, ...)	Looks up a value in a table based on conditions.	LOOKUPVALUE(Product[Price], Product[ProductID], Sales[ProductID])
VALUES	VALUES(Column)	Returns unique values from a column.	VALUES(Sales[Region])
SELECTEDVALUE	SELECTEDVALUE(Column, Alt)	Returns selected value from context or alternate result.	SELECTEDVALUE(Sales[Region], "All")
HASONEVALUE	HASONEVALUE(Column)	Checks if only one value is selected.	HASONEVALUE(Sales[Region])
HASONEFILTER	HASONEFILTER(Column)	Checks if column has one filter.	HASONEFILTER(Sales[Region])
ISFILTERED	ISFILTERED(Column)	Checks if column is filtered (in any way).	ISFILTERED(Sales[Region])

EXAMPLE - RELATED

Get Customer Name in Sales Table

Customer Name = RELATED(Customer[Customer Name])



★ KEY TAKEAWAY

- ✓ Relationships + DAX functions = Powerful Analytics.
- ✓ They help fetch values, apply filters, and create smart reports.



DID YOU KNOW?

is not just formulas.
It is a powerful language
that thinks like a business
analyst!



PRO TIP

Don't just learn DAX functions.
Understand the context (Row vs Filter)
and apply them to real business problems.
Practice. Build. Analyze. Grow!



“ Master DAX today,
become the Analyst your
company depends on tomorrow. ”

- Tech With Shivansh



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8. TIME INTELLIGENCE FUNCTIONS

Function	Syntax	Description	Example	Use Case
DATEADD	DATEADD(Dates[Date], Interval, Number)	Shifts dates by a specified interval.	DATEADD('Date'[Date], YEAR, -1)	Previous Year Comparison
SAMEPERIODLASTYEAR	SAMEPERIODLASTYEAR(Dates[Date])	Returns same period in the previous year.	SAMEPERIODLASTYEAR('Date'[Date])	YoY Analysis
PARALLELPERIOD	PARALLELPERIOD(Dates[Date], Interval, Number)	Shifts dates by given interval (Number of periods).	PARALLELPERIOD('Date'[Date], MONTH, -1)	Previous Month Comparison
TOTALYTD	TOTALYTD(Expression, Dates[Date])	Cumulative total from start of year to current date.	TOTALYTD([Total Sales], 'Date'[Date])	YTD Sales
DATESYTD	DATESYTD(Dates[Date])	Returns all dates from start of year to current date.	DATESYTD('Date'[Date])	Used in custom calculations
TOTALMTD	TOTALMTD(Expression, Dates[Date])	Cumulative total for current month.	TOTALMTD([Total Sales], 'Date'[Date])	MTD Sales
DATESMTD	DATESMTD(Dates[Date])	Returns all dates in current month.	DATESMTD('Date'[Date])	Month to Date Analysis
TOTALQTD	TOTALQTD(Expression, Dates[Date])	Cumulative total for current quarter.	TOTALQTD([Total Sales], 'Date'[Date])	QTD Sales
DATESQTD	DATESQTD(Dates[Date])	Returns all dates in current quarter.	DATESQTD('Date'[Date])	Quarter Analysis
PREVIOUSMONTH	PREVIOUSMONTH(Dates[Date])	Returns all dates in previous month.	PREVIOUSMONTH('Date'[Date])	Previous Month Trend
NEXTMONTH	NEXTMONTH(Dates[Date])	Returns all dates in next month.	NEXTMONTH('Date'[Date])	Future Analysis
DATESBETWEEN	DATESBETWEEN(Dates[Date], StartDate, EndDate)	Returns dates between start and end date.	DATESBETWEEN('Date'[Date], DATE(2024,1,1), DATE(2024,12,31))	Custom Range Analysis

EXAMPLE – YoY SALES COMPARISON

Current Year Sales = [Total Sales]
Previous Year Sales =
CALCULATE(
[Total Sales],
SAMEPERIODLASTYEAR('Date'[Date])
)
YoY Growth % =
DIVIDE(
[Current Year Sales] - [Previous Year Sales],
[Previous Year Sales], 0
)



★ KEY TAKEAWAYS

- ✓ Time Intelligence functions are the heart of trend analysis.
- ✓ They help compare data across different time periods.
- ✓ Essential for dashboards and KPIs.
- ✓ Used heavily in Finance, Sales, Inventory & HR analytics.



9. TABLE & INFORMATION FUNCTIONS

Function	Syntax	Description	Example	Use Case
VALUES	VALUES(Table[Column])	Returns a table of unique values.	VALUES(Customer[City])	Get unique list for slicers
DISTINCT	DISTINCT(Table[Column])	Returns unique values (remove duplicates).	DISTINCT(Product[Category])	Unique categories
SELECTEDVALUE	SELECTEDVALUE(Column, Alternate Result)	Returns selected value if only one value is selected.	SELECTEDVALUE('Date'[Year], "All Years")	Show single selected value
HASONEVALUE	HASONEVALUE(Column)	Checks if only one value is selected.	HASONEVALUE(Customer[Segment])	Conditional logic
HASONEFILTER	HASONEFILTER(Column)	Checks if column has a filter.	HASONEFILTER('Date'[Year])	Filter detection
ISFILTERED	ISFILTERED(Column)	Checks if column is filtered.	ISFILTERED(Customer[City])	Dynamic titles / logic
COUNTROWS	COUNTROWS(Table)	Counts number of rows in a table.	COUNTROWS(Sales)	Total Records
SUMMARIZE	SUMMARIZE(Table, Column1, Column2, ...)	Groups data and creates a summary table.	SUMMARIZE(Sales, Customer[City], Product[Category])	Custom summary
ADDCOLUMNS	ADDCOLUMNS(Table, Name, Expression)	Adds calculated columns to a table.	ADDCOLUMNS(Sales, "Total", [Total Sales])	Extend tables
GENERATE	GENERATE(Table1, Table2)	Returns combined rows from two tables.	GENERATE(Customer, RELATEDTABLE(Sales))	Advanced modeling
CROSSJOIN	CROSSJOIN(Table1, Table2)	Returns Cartesian product of two tables.	CROSSJOIN(VALUES('Date'[Year]), VALUES(Product[Category]))	All combinations
UNION	UNION(Table1, Table2)	Appends rows from tables.	UNION(TableA, TableB)	Combine similar tables
INTERSECT	INTERSECT(Table1, Table2)	Returns common rows from tables.	INTERSECT(TableA, TableB)	Compare tables
EXCEPT	EXCEPT(Table1, Table2)	Returns rows from Table1 not in Table2.	EXCEPT(TableA, TableB)	Find differences

EXAMPLE – SELECTEDVALUE

Selected Year =
SELECTEDVALUE('Date'[Year], "All Years")

Result =
IF(
HASONEVALUE('Date'[Year]),
"You selected: " & [Selected Year],
"Multiple Years Selected"
)

BUSINESS USE CASES

- ✓ Create dynamic titles in dashboards.
- ✓ Build slicers with clean values.
- ✓ Use summarization for management reports.
- ✓ Compare tables for data quality.
- ✓ Powerful in advanced analytics models.



⚙️ PRACTICAL EXAMPLE – BUILDING KPI USING DAX

1. Total Sales Total Sales = SUM(Sales[Amount]) → Total revenue from all transactions	2. Total Profit Total Profit = SUM(Sales[Profit]) → Total profit earned by the business	3. Profit Margin % Profit Margin % = DIVIDE([Total Profit], [Total Sales], 0) → Profit as a % of Sales	4. YoY Growth % YoY Growth % = DIVIDE([Total Sales] - [PY Sales], [PY Sales], 0) → Growth compared to last year	5. Running Total Sales Running Total Sales = CALCULATE([Total Sales], FILTER(ALL('Date'[Date]), 'Date'[Date] <= MAX('Date'[Date]))) → Cumulative sales till date
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💡 PRO TIP
Start simple, Understand context, Practice daily, Build real dashboards, Master DAX!

DID YOU KNOW?
DAX is a hybrid language. It understands both functions and context. That's why it's powerful!

BENEFITS OF DAX

- ✓ Speeds up complex calculations
- ✓ Works on millions of rows easily
- ✓ Reduces dependency on Excel formulas
- ✓ Improves report performance
- ✓ Essential skill for Data Analyst & BI roles

“Data tells a story, DAX helps you explain it in the most powerful way.”
– Tech With Shivansh

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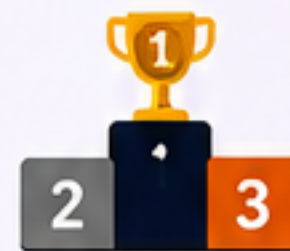
10. RANKING & ORDERING FUNCTIONS

Function	Syntax	Description	Example	Use Case
RANKX	RANKX(Table, Expression, [Value], [Order], [Ties])	Ranks values in a category.	Rank = RANKX(ALL(Customer), [Total Sales])	Top Customers by Sales
TOPN	TOPN(N, Table, OrderBy_Expression, [Order])	Returns top N rows.	Top 5 Products = TOPN(5, Product, [Total Sales], DESC)	Top 5 Products
MINX / MAXX	MINX(Table, Expression) MAXX(Table, Expression)	Returns min / max of expression.	Max Order Value = MAXX(Orders, Orders[Amount])	Min / Max transactions
EARLIER	EARLIER(Column, [Number])	Refers to previous row context.	Calculated Column with nested row context	Advanced calculated columns
GENERATE	GENERATE(Table1, Table2)	Returns Cartesian product.	GENERATE(Customer, RELATEDTABLE(Sales))	Expand related rows
SUMMARIZE	SUMMARIZE(Table, GroupBy, "Name", Expression, ...)	Groups data & creates summary table.	SUMMARIZE(Sales, Customer[City], "Total Sales", SUM(Sales[Amount]))	Summary reports
ADDCOLUMNS	ADDCOLUMNS(Table, Name, Expression, ...)	Adds calculated columns.	ADDCOLUMNS(Product, "Total Sales", [Total Sales])	Extend tables
CROSSJOIN	CROSSJOIN(Table1, Table2)	Cartesian product of two tables.	CROSSJOIN(Date, Product)	All combinations
UNION	UNION(Table1, Table2)	Appends rows from tables.	UNION(Previous Year, Current Year)	Combine similar tables
INTERSECT	INTERSECT(Table1, Table2)	Returns common rows.	INTERSECT(TableA, TableB)	Common records
EXCEPT	EXCEPT(Table1, Table2)	Returns rows from Table1 not in Table2.	EXCEPT(TableA, TableB)	Find differences

EXAMPLE – RANKX

Rank Customers by Total Sales

Customer Rank =
RANKX(
 ALL(Customer),
 [Total Sales],
 ,
 DESC,
 DENSE
)



★ KEY TAKEAWAYS

- ✓ Ranking functions help in leaderboards.
- ✓ Use TOPN for Top N analysis.
- ✓ SUMMARIZE & ADDCOLUMNS are powerful for custom tables.
- ✓ UNION, INTERSECT, EXCEPT help in data comparison.



11. TIME INTELLIGENCE FUNCTIONS

Function	Syntax	Description	Example	Use Case
DATEADD	DATEADD(Dates, Interval, Number)	Shifts dates by interval.	DATEADD('Date'[Date], YEAR, -1)	Previous Year Comparison
SAMEPERIODLASTYEAR	SAMEPERIODLASTYEAR(Dates)	Same period in previous year.	SAMEPERIODLASTYEAR('Date'[Date])	YoY Analysis
PARALLELPERIOD	PARALLELPERIOD(Dates, Interval, Number)	Shifts dates by given interval.	PARALLELPERIOD('Date'[Date], MONTH, -1)	Previous Month Comparison
TOTALYTD	TOTALYTD(Expression, Dates)	Cumulative total from start of year.	TOTALYTD([Total Sales], 'Date'[Date])	YTD Sales
DATESYTD	DATESYTD(Dates)	Returns all dates YTD.	DATESYTD('Date'[Date])	Used in custom calculations
TOTALMTD	TOTALMTD(Expression, Dates)	Cumulative total MTD.	TOTALMTD([Total Sales], 'Date'[Date])	MTD Sales
DATESMTD	DATESMTD(Dates)	Returns all dates MTD.	DATESMTD('Date'[Date])	MTD Analysis
TOTALQTD	TOTALQTD(Expression, Dates)	Cumulative total QTD.	TOTALQTD([Total Sales], 'Date'[Date])	QTD Sales
DATESQTD	DATESQTD(Dates)	Returns all dates QTD.	DATESQTD('Date'[Date])	QTD Analysis
PREVIOUSMONTH	PREVIOUSMONTH(Dates)	Returns dates in previous month.	PREVIOUSMONTH('Date'[Date])	Previous Month Trend
NEXTMONTH	NEXTMONTH(Dates)	Returns dates in next month.	NEXTMONTH('Date'[Date])	Future Analysis
DATESBETWEEN	DATESBETWEEN(Dates, StartDate, EndDate)	Returns dates between range.	DATESBETWEEN('Date'[Date], DATE(2024,1,1), DATE(2024,12,31))	Custom Range Analysis

EXAMPLE – YTD SALES

YTD Sales =
TOTALYTD(
 [Total Sales],
 'Date'[Date]
)



BUSINESS USE CASES

- ✓ Compare sales across years.
- ✓ Track monthly, quarterly, yearly performance.
- ✓ Build trends and seasonality reports.
- ✓ Essential for Financial Reporting, Budgeting & Forecasting.



12. COMMON KPI MEASURES USING DAX

1. TOTAL REVENUE Total Revenue = SUM(Sales[Amount])	2. TOTAL PROFIT Total Profit = SUM(Sales[Profit])	3. PROFIT MARGIN % Profit Margin % = DIVIDE([Total Profit], [Total Revenue], 0)	4. YoY GROWTH % YoY Growth % = DIVIDE([Total Sales] - [PY Sales], [PY Sales], 0)	5. MoM GROWTH % MoM Growth % = DIVIDE([Total Sales] - [PM Sales], [PM Sales], 0)	6. RUNNING TOTAL Running Total Sales = CALCULATE([Total Sales], FILTER(ALL('Date'[Date]), 'Date'[Date] <= MAX('Date'[Date])))
7. CUSTOMER COUNT Customer Count = DISTINCTCOUNT(Customer[CustomerID])	8. NEW CUSTOMERS New Customers = CALCULATE(DISTINCTCOUNT(Customer[CustomerID]), Customer[Is New] = "Yes")	9. AVERAGE ORDER VALUE Avg Order Value = DIVIDE([Total Revenue], DISTINCTCOUNT(Orders[OrderID]), 0)	10. CONTRIBUTION % Contribution % = DIVIDE([Total Revenue], CALCULATE([Total Revenue], ALL(Product)), 0)	11. INVENTORY TURNOVER Inventory Turnover = DIVIDE([COGS], AVERAGE(Inventory[Stock Value]), 0)	

KPI INSIGHT

KPIs tell the story of your business. Build the right measures and your dashboard will speak the truth about performance!

PRO TIP

Always build reusable measures. Use Variables in DAX for better performance and clean code.

“

Good data + Good DAX = Powerful Insights!”

– Tech With Shivansh

”

13. TABLE MANIPULATION FUNCTIONS

Function	Syntax	Description	Example	Use Case
SUMMARIZE	SUMMARIZE(Table, GroupBy_Column, "Name", Expression, ...)	Groups data and calculates expressions.	SUMMARIZE(Sales, Customer[City], "Total Sales", SUM(Sales[Amount]))	Summary reports
ADDCOLUMNS	ADDCOLUMNS(Table, "Name", Expression, ...)	Adds calculated columns to an existing table.	ADDCOLUMNS(Customer, "Total Sales", [Total Sales])	Extend tables
SELECTCOLUMNS	SELECTCOLUMNS(Table, "Name", Expression, ...)	Returns a table with selected/renamed columns.	SELECTCOLUMNS(Product, "Product Name", Product[Name])	Pick required columns
CALCULATETABLE	CALCULATETABLE(Table, Filter1, Filter2, ...)	Returns a table after applying filters.	CALCULATETABLE(Sales, Sales[Amount] > 1000)	Filtered table
FILTER	FILTER(Table, Condition)	Returns a table filtered by condition.	FILTER(Sales, Sales[Amount] > 500)	Row filtering
GENERATE	GENERATE(Table1, Table2)	Returns Cartesian product of tables.	GENERATE(Customer, RELATEDTABLE(Sales))	Expand related rows
GENERATEALL	GENERATEALL(Table1, Table2)	Same as GENERATE but keeps unmatched rows.	GENERATEALL(Customer, Sales)	Include unmatched
CROSSJOIN	CROSSJOIN(Table1, Table2)	Cartesian product of two tables.	CROSSJOIN(VALUES(Date[Year]), VALUES(Product[Category]))	All combinations
UNION	UNION(Table1, Table2)	Combines rows from tables.	UNION(TopCustomers, NewCustomers)	Append tables
INTERSECT	INTERSECT(Table1, Table2)	Returns common rows.	INTERSECT(TableA, TableB)	Common records
EXCEPT	EXCEPT(Table1, Table2)	Returns rows in Table1 not in Table2.	EXCEPT(AllCustomers, InactiveCustomers)	Find differences
TREATAS	TREATAS(TableExpression, Column1, Column2, ...)	Applies table as filter to unrelated table.	TREATAS(VALUES(Geo[State]), Sales[State])	Virtual relationship

EXAMPLE - SUMMARIZE + ADDCOLUMNS

Get Total Sales by City with Average Order Value

Result Table =

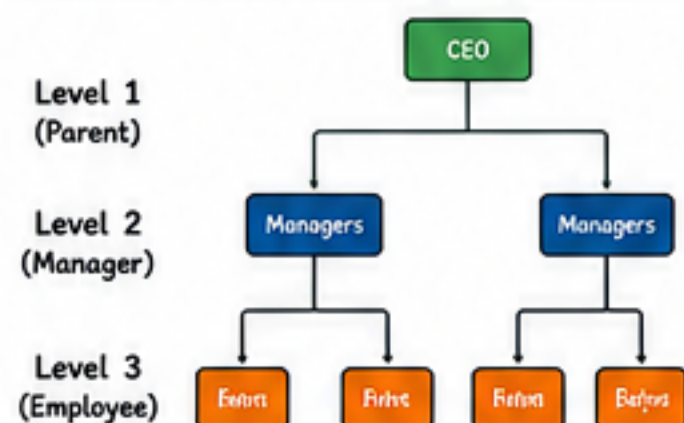
```
ADDCOLUMNS(
    SUMMARIZE(Sales, Customer[City]),
    "Total Sales", SUM(Sales[Amount]),
    "Avg Order Value",
    DIVIDE(SUM(Sales[Amount]), COUNTROWS(Sales))
)
```

City	Total Sales	Avg Order Value
Delhi	12,45,000	2,850
Mumbai	9,85,000	2,625
Bangalore	8,75,000	2,430
...	...	...

★ KEY TAKEAWAYS

- ✓ Table functions help shape and transform data.
- ✓ Essential for advanced data modeling.
- ✓ Help create virtual tables for complex analysis.
- ✓ Used widely in Power Pivot & Power BI.

14. PARENT & CHILD, HIERARCHY FUNCTIONS



Function	Syntax	Description	Example
PATH	PATH(ParentID, ChildID)	Returns path from parent to child.	PATH(Employee[ManagerID], Employee[EmployeeID])
PATHITEM	PATHITEM(Path, Position, Type)	Returns value from specific position.	PATHITEM([Path], 1, TEXT)
PATHLENGTH	PATHLENGTH(Path)	Returns number of levels in path.	PATHLENGTH([Path])
PATHCONTAINS	PATHCONTAINS(Path, Value)	Checks if path contains a value.	PATHCONTAINS([Path], 102)
PATHITEMREVERSE	PATHITEMREVERSE(Path, Position, Type)	Returns value from end position.	PATHITEMREVERSE([Path], 1, TEXT)
ISINSCOPE	ISINSCOPE(Column)	Checks if column is in the current hierarchy.	ISINSCOPE(Date[Year])

EXAMPLE - ORGANIZATIONAL HIERARCHY

Get Employee Level in Organization

Employee Level = PATHLENGTH(Employee[Path])

Result:

CEO - Level 1
Manager - Level 2
Team Lead - Level 3
Employee - Level 4



BUSINESS USE CASES

- ✓ Analyze hierarchy based organizations.
- ✓ Calculate employee levels dynamically.
- ✓ Build drill-down dashboards.
- ✓ Used in HR, Sales Teams, Reporting Structures.

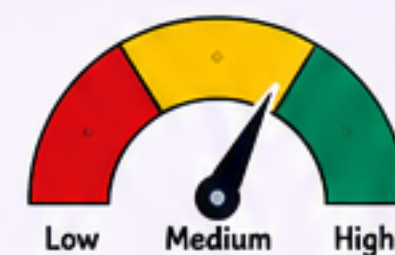
15. STATISTICAL & MATHEMATICAL FUNCTIONS

Function	Syntax	Description	Example	Use Case
ABS	ABS(Number)	Returns absolute value.	ABS(-250)	Remove negative
ROUND	ROUND(Number, Digits)	Rounds number to given digits.	ROUND(123.456, 2)	Round values
INT	INT(Number)	Rounds down to nearest integer.	INT(9.87)	Whole number
MOD	MOD(Number, Divisor)	Returns remainder after division.	MOD(10, 3)	Buckets, patterns
POWER	POWER(Number, Power)	Returns number raised to power.	POWER(2, 3)	Growth calc
SQRT	SQRT(Number)	Returns square root of number.	SQRT(16)	Math calc
LOG	LOG(Number, Base)	Returns logarithm of number.	LOG(100, 10)	Data scaling
SIGN	SIGN(Number)	Returns -1, 0 or 1 based on sign.	SIGN(-5)	Direction check

EXAMPLE - PROFIT MARGIN BUCKET

Margin Bucket =

```
SWITCH(TRUE(),
    [Profit Margin %] >= 0.20, "High",
    [Profit Margin %] >= 0.10, "Medium",
    [Profit Margin %] >= 0, "Low",
    "Loss"
)
```



💡 KEY TAKEAWAYS

- ✓ Mathematical functions are important for KPIs.
- ✓ Use MOD for bucketing and grouping.
- ✓ ROUND, INT, ABS are widely used in reports.
- ✓ Helps in advanced calculations and data preparation.

16. ADVANCED DAX PATTERNS

1. RUNNING TOTAL

```
Running Sales =
CALCULATE(
    [Total Sales],
    FILTER(ALL(Date[Date]),
        Date[Date] <= MAX(Date[Date])
    )
)
```



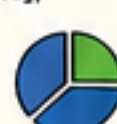
2. MOVING AVERAGE (3 MONTHS)

```
Moving Avg 3M =
AVERAGEX(
    DATESINPERIOD(
        Date[Date],
        MAX(Date[Date]),
        -3, MONTH
    ),
    [Total Sales]
)
```



3. PERCENTAGE OF TOTAL

```
% of Total Sales =
DIVIDE(
    [Total Sales],
    CALCULATE([Total Sales],
        ALL(Sales))
)
```



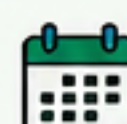
4. TOP N ITEMS

```
Top 5 Products Sales =
TOPN(5, VALUES(Product[Product Name]),
    [Total Sales], DESC
)
```



5. ROLLING 12 MONTHS SALES

```
Rolling 12M Sales =
CALCULATE(
    [Total Sales],
    DATESINPERIOD(
        Date[Date],
        MAX(Date[Date]),
        -12, MONTH
    )
)
```



💡 DID YOU KNOW?

DAX is not just about formulas, it's about understanding business logic and data behavior!

🚀 PRO TIP

Always start with simple measures, understand filter context, and then move to advanced DAX patterns.

“ Learn DAX, Think Like Analyst, Solve Like Expert! ”

- Tech With Shivansh

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17. DATE & TIME HELPER FUNCTIONS

Function	Syntax	Description	Example	Use Case
YEAR	YEAR(Date)	Returns year from a date.	YEAR('2024-05-18')	Extract year
MONTH	MONTH(Date)	Returns month number.	MONTH('2024-05-18')	Extract month
DAY	DAY(Date)	Returns day number.	DAY('2024-05-18')	Extract day
WEEKDAY	WEEKDAY(Date, [ReturnType])	Returns day of week.	WEEKDAY(TODAY(), 2)	Day of week
MONTHNAME	MONTHNAME(MonthNumber, [ReturnType])	Returns month name.	MONTHNAME(5)	Month name
FORMAT	FORMAT(Date, "Format Text")	Formats date as text.	FORMAT(TODAY(), "MMM YYYY")	Custom format
TODAY	TODAY()	Returns current date.	TODAY()	Current date
NOW	NOW()	Returns current date & time.	NOW()	Current date time
EOMONTH	EOMONTH(StartDate, Months)	Returns last day of month.	EOMONTH(TODAY(), 0)	Month end
EDATE	EDATE(StartDate, Months)	Returns date after months.	EDATE(TODAY(), -3)	Past/future date
DATEDIFF	DATEDIFF(Date1, Date2, Interval)	Difference between dates.	DATEDIFF(Date[Date], TODAY(), DAY)	Age, tenure, etc.
NETWORKDAYS	NETWORKDAYS(Start, End, [Holidays])	Working days between dates.	NETWORKDAYS('2024-01-01', '2024-01-31')	Workday calc
DATEVALUE	DATEVALUE(Text)	Converts text to date.	DATEVALUE('18-05-2024')	Text to date
TIMEVALUE	TIMEVALUE(Text)	Converts text to time.	TIMEVALUE('09:30 AM')	Text to time

EXAMPLE – MONTHLY SALES TREND

Monthly Sales =
CALCULATE(
[Total Sales],
DATESMTD('Date'[Date])
)



BUSINESS USE CASES

- ✓ Trend analysis (MTD, QTD, YTD).
- ✓ Compare dates across periods.
- ✓ Build Month over Month dashboards.
- ✓ Create rolling periods & forecasts.
- ✓ Essential for Financial & Sales reporting.



★ KEY TAKEAWAYS

- ✓ Date functions help in time analysis.
- ✓ Used widely in KPIs & dashboards.
- ✓ Combine with CALCULATE for powerful results.



18. ALL DAX FUNCTIONS QUICK REFERENCE

Σ AGGREGATION FUNCTIONS

- SUM, SUMX
- AVERAGE, AVERAGEX
- MIN, MAX
- COUNT, COUNTA, COUNTROWS
- DISTINCTCOUNT
- DIVIDE
- COALESCE
- STDEVX.P, STDEVX.S, VAR.P, VAR.S

🧠 LOGICAL FUNCTIONS

- IF
- SWITCH
- AND, OR, NOT
- TRUE, FALSE
- BLANK, ISBLANK
- ISERROR
- IFERROR

🔎 FILTER MODIFIER FUNCTIONS

- CALCULATE
- FILTER
- ALL
- ALLEXCEPT
- ALLSELECTED
- REMOVEFILTERS
- KEEPFILTERS
- CROSSFILTER
- USERELATIONSHIP

🔗 RELATIONSHIP FUNCTIONS

- RELATED
- RELATEDTABLE
- LOOKUPVALUE
- VALUES
- SELECTEDVALUE
- HASONESVALUE
- ISFILTERED
- HASONEFILTER

🕒 TIME INTELLIGENCE FUNCTIONS

- DATEADD
- SAMEPERIODLASTYEAR
- PARALLELPERIOD
- TOTALYTD, DATESYTD
- TOTALQTD, DATESQTD
- TOTALMTD, DATESMTD
- PREVIOUSMONTH
- NEXTMONTH
- DATESBETWEEN
- EOMONTH, EDATE

🏆 RANKING & ORDERING

- RANKX
- TOPN
- MINX, MAXX
- EARLIER
- GENERATE

📅 TABLE MANIPULATION

- SUMMARIZE
- ADDCOLUMNS
- SELECTCOLUMNS
- CALCULATETABLE
- GENERATEALL
- CROSSJOIN
- UNION
- INTERSECT
- EXCEPT

📊 STATISTICAL FUNCTIONS

- ABS
- ROUND, ROUNDUP, ROUNDDOWN
- INT
- MOD
- POWER
- SQRT
- LOG, LOG10
- SIGN

ℹ️ INFORMATION FUNCTIONS

- ISNUMBER
- ISTEXT
- ISLOGICAL
- ISBLANK
- DATATYPE
- ERROR
- ISINSCOPE

🔧 OTHER HELPFUL FUNCTIONS

- CONCATENATE
- CONCATENATEX
- TREATAS
- PATH, PATHITEM
- PATHLENGTH
- PATHCONTAINS
- PATHITEMREVERSE

COMPLETE DAX WORKFLOW



DAX TIPS & BEST PRACTICES

- ✓ Use Measures, not Calculated Columns, for aggregations.
- ✓ Understand Row Context vs Filter Context.
- ✓ Use CALCULATE wisely to control filters.
- ✓ Use Variables (VAR) for better performance.
- ✓ Keep data model clean & relationships optimal.
- ✓ Test measures at each step.
- ✓ Follow naming conventions for measures.



DID YOU KNOW?

DAX is the brain behind
Power Pivot & Power BI.
Master DAX = Master Analytics!



PRO TIP

Practice DAX every day.
The more you write,
the more you will Master!



“ DAX turns your data
into Decisions.
Learn. Practice. Grow.
– Tech With Shivansh

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★ 19. INTERVIEW QUESTIONS & ANSWERS (DAX FOR ANALYTICS)

#	QUESTION	ANSWER (SHORT)
16	What is the use of DIVIDE()?	Performs division and handles divide by zero errors.
17	What is SELECTEDVALUE()?	Returns the selected value if one value is selected from a column, else returns alternate result.
18	How to calculate Running Total?	Use CALCULATE with FILTER and <= condition on date.
19	How to calculate Moving Average?	Use AVERAGEX with DATESINPERIOD.
20	What is the use of KEEPFILTERS()?	Adds filters without removing existing filters.
21	What is RANKX()?	Ranks data based on a given expression.
22	How do you find Top N items?	Use TOPN(N, Table, [Measure], DESC).
23	What is ADDCOLUMNS()?	Adds calculated columns to a table.
24	What is SUMMARIZE()?	Creates summary table with grouping and aggregated columns.
25	What is CROSSFILTER()?	Changes the filter direction between two tables.
26	What happens if relationship is inactive?	It is ignored in calculation unless activated using USERELATIONSHIP().
27	What is EARLIER()?	Used in calculated column to access the row context of outer loop.
28	How do you calculate % of Total?	% of Total = DIVIDE([Value], CALCULATE([Value], ALL(Table)))
29	What is the default aggregation in DAX?	SUM
30	How can you optimize DAX performance?	Use proper data model, avoid calculated columns, use variables, minimize FILTER in measures.

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